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C. Fast Sum

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: `sort`, `open`, `file`, `creat`(, `fstream`, `thread`, `process`, `system`(, `exec`(

Your friend is trying to solve the following problem. You are given T test cases. For each test case, you are given an integer N . You have to find out the summation of 1 to N . More formally, your friend has to calculate

$$\sum_{x=1}^N x$$

Your friend wrote the following code in Python to solve it:

```
T = int(input())

for _ in range(T):
    N = int(input())
    sum = 0
    for i in range(1, N + 1):
        sum += i
    print(sum)
```

Same code in Java:

```
import java.util.Scanner;

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int T = sc.nextInt();
        for (int t = 0; t < T; t++) {
            int N = sc.nextInt();
            int sum = 0;
            for (int i = 1; i <= N; i++)
                sum += i;
            System.out.println(sum);
        }
    }
}
```

However, the code is not passing the online judge due to some unknown errors for large values of N .

Since you are currently studying CSE221 and have learned about time complexity, help your friend come up with a more efficient solution.

CSE221 Assignment 01 Summer 2026

Contest is running

2 days

Contestant



→ Submit?

Language: Java 21 64bit 

Choose file: Choose File No file chosen



Success!



Submit

 Input

The first line contains a single integer T ($1 \leq T \leq 10^4$) — the number of test cases.

The next T lines each contain a single integer N ($1 \leq N \leq 10^6$).

Output

For each test case, print a single integer — the summation from 1 to N .

Example

input	Copy
5 2 5 10 12 100	
output	Copy
3 15 55 78 5050	

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